# The Swap-Refresh Gadgetization

symmetric-ports revision

local_mixing — design note

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Redesign of the single-carrier production gadgetizer (the `gen_sandwich_gadget` sandwich path). The aggregate-Gray fold is replaced by a per-gate *mask swap-with-refresh*; both ports carry an independently drawn *junk-half zero-slice guard*; the tail is leak-hygienic; and the composite is *reverse-honest*: the same artifact evaluates $C(x)$ forward and $D^{-1}(a)$ backward, each on its own zero slice — the gadget-level mirror of the sandwich's symmetry.

## 1 Goal

No GF(2)-linear equation should hold in which some variables are wire segments of the source (sandwich) circuit and others are wire segments of the gadgetized circuit — except through the public port boundary, which any correct implementation exhibits. The prior construction failed this three measured ways: the endpoint cancellation (time-invariant masks cancel in every fold's before/after XOR; `flip_match`: 100% of linear source gates), the aggregate-Gray operand witness ($B = c_b \oplus u^- \oplus u^+ \oplus \kappa$), and the one-sided zero-slice port. Adopted during the redesign: the construction must be *symmetric under reversal*, and the stream must remain *predominantly g57/CNOT* for phase-A digestion.

## 2 The construction

Layout under `production_single()` (`PROD_SWAP=0 PROD_CLOSE_SLICE=0` restores the prior Gray stream byte-for-byte):

[$P$ junk-guard] [$W_0$ fill] [inject] [folds + swap/reloc churn] [route home] [strip ALL] [$F'$ fill] [$Q$ junk-guard]

- **Per-gate mask swap-with-refresh**: at every fold the target and one random control each retire one base-degree monomial and gain a fresh draw (a verbatim move leaks by GF(2) conservation, a polarity re-roll by low-degree difference). The target-side inject is strictly interior to the fold's fragment stream; a target-stable commuting shuffle preserves per-wire write order; every relocation also refreshes one monomial of each moved value. Gray modes are declined.
- **Symmetric ports**: $P$ and $Q$ are independent draws of one generator — identity exactly on the zero band slice, every nonzero slice perturbs the data, targets confined to the LOW (forward-junk) data half. $Q$ fires forward into the junk half; $P^{-1}$ fires backward into the same half; neither touches the half where the live payload of its direction emerges.

- **Reverse honesty (the telescope lemma)**: a fold reading value $v$ compensates $v$'s mask fragments against the carrier's reverse-time content — the XOR of the emissions *after* the fold — which telescopes to $v$'s fold-time slots exactly when the final registry is stripped. So *every* value's registry is discharged at the tail; keeping the junk half masked (an earlier revision) corrupted $D^{-1}$. Verified bit-sliced at every size, forward and reverse, plus a permanent reverse verify in `gen_sandwich_gadget`.

- **Tail hygiene** (each item closed a measured `flip_match` class): route-home precedes the strip and carries the band-placement map; the constant-discharge helpers are live band *variables* via `loc` (the raw band wire range holds displaced carrier states); both band fills source from the low data half; epoch refills split their channels — the linear pivot is band-sourced only (a carrier-sourced pivot linearly copies a masked payload-era state into the band, the measured exact leak, since the low half is the payload's birthplace mid-circuit) while the product terms readmit carriers at `refill_data`% (degree-2 in the values: nothing linearly peelable, refill clusters read across the band/carrier partition, genuinely new algebra enters the band, no rank drift); the ladder borrows band variables, never live carriers.

## 3 Vocabulary: the ladder and the g57 share

The expanded fold's arity-2 product fragments are wide; fmix's store speaks g57/CNOT. The selective ladder re-spells them; its scratch is the band pool (live-carrier borrows exposed data states — the measured cap-4 leak). Menu at $n = 128$, production parameters:

| ladder | gates | g57 | CNOT | conj-2 | wide | vs Gray |
|---|---|---|---|---|---|---|
| cap 0 | 882,768 | 28.7% | 22.7% | 0.2% | 48.4% | 0.68× |
| **cap 3** | **1,687,450** | **45.4%** | **36.2%** | 0.1% | 18.3% | **1.31**× |
| cap 4 | 3,297,951 | 34.7% | 26.6% | 35.4% | 3.3% | 2.55× |

Cap 4's extra narrowness arrives as store-weak plain conj-2 gates at twice cap 3's size, so cap 3 is the default (81.6% pure g57+CNOT); `PROD_LADDER_CAP` is the campaign lever. The old Gray build was 97.6% g57+CNOT at 1,292,677 gates — but carries the operand witness.

## 4 Verification status ($n = 128$, fresh random seeds)

Function forward *and* reverse: bit-sliced verifies pass; 318/318 tests. Inputs↔outputs: affine rank of $\{1, x, y\}$ full (257/257). `segment_deduce` (degree 1 and restricted degree-2 windows, both directions, fresh-sample verified): zero interior-cut equations. Affine-predictor heatmap vs $C$: flat at the 0.5 ceiling, port corners only, indistinguishable from the Gray control plate. `flip_match` (kwin 12/30/60, 8192 samples): body and nonlinear classes at zero in both orientations. Residual: 0–2 exact *seam windows* per build (seed-dependent; 6/2/0/5 matched source gates over four seeds, all $\geq 99.4\%$ depth) formed by mid-group cuts across the route/strip seam, plus occasional port-boundary coincidences (early $W_0$ / late $F'$ fill products over public port values equal to early-C / late-D gate deltas). The seam is the region phase-A mixing rewrites most heavily; driving the gadget-level flicker to zero at every seed would need per-wire alternation of fresh material through the seam — an open refinement.